

Scale or Shape? Decomposing NTK Movement During Grokking

Group G2

Zain Al-Saffi (s4800836) · Jasper Chong (s4883420) · Aaron D’Mello (s4890257)

Zac Kienzle (s4803349) · Samuel Rieger (s4904959)

Abstract

FiXme Fatal: Abstract, written last from the finished sections.

1 Introduction

Grokking is the delayed generalisation that Power et al. (2022) found on small algorithmic datasets: validation accuracy rises from chance long after the training data have been fitted. Modular addition learned by a two-layer perceptron is a minimal instance. Full-batch gradient descent on the squared error groks there with no explicit regularisation (Gromov, 2023), and Kumar et al. (2024) use this setting, with no weight decay, as a baseline.

One account holds that grokking is a transition from lazy training, in which the network behaves as a kernel method with its initial tangent kernel, to rich training, in which the kernel adapts to the task (Kumar et al., 2024). The evidence that the kernel changes is a distance: Mohamadi et al. (2024) observe that the empirical NTK stays close to its initial value until the training data are fitted and moves far from it afterwards. A distance does not say how the kernel changed. Shan and Bordelon (2022) find that the tangent kernel grows in Frobenius norm during training, as Baratin et al. (2021) report, but that kernel gradient descent with the initial kernel scaled up to the amplitude of the final one still trains more slowly than the network, so that changes to the structure of the kernel, and not its amplitude alone, account for the faster training.

The distinction matters for accounts of what sets the grokking time. The result cited for weight decay forcing the NTK to change (Kumar et al., 2024) is that of Lewkowycz and Gur-Ari (2021), in which the kernel only rescales, and recent accounts time grokking by the weight norm (Khanh et al., 2026), the scale of the logits (Khanh, 2026) and the rate of weight decay (Kim, 2026). None of them measures the shape of the kernel.

This report separates the two with published measures of the tangent kernel (section 3.3): its scale S_t , the Frobenius norm relative to initialisation; its shape change R_t , the kernel distance of Fort et al. (2020), which compares kernels in a scale-invariant manner; its relative variation D_t of Geiger et al. (2020); and its alignment A_t with the labels. We measure these statistics and the kernel’s spectrum along training on the modular-addition baseline of Kumar et al. (2024), and ask which of scale and shape moves before and while the network groks, and whether the times at which it memorises and groks follow either. **FiXme Fatal: State the findings and the contributions once the grid has run.**

2 Related Work

Grokking. Power et al. (2022) report that on small algorithmic datasets, long after a network has overfitted its training data, validation accuracy can rise from chance towards perfect generalisation, and that weight decay is particularly effective at improving generalisation there. Nanda et al. (2023) reverse-engineer the algorithm a small transformer learns for modular addition, which turns addition into rotation about a circle through discrete Fourier transforms, and use it to split training into memorisation, circuit formation and clean-up. For the two-layer perceptron we study, Gromov (2023) gives closed-form weights that solve modular addition when the activation is quadratic, and observes grokking under full-batch gradient descent with no explicit regularisation. Liu, Michaud, and Tegmark (2023) trace grokking to a mismatch between the training and test losses as functions of the weight norm.

Lazy and rich training. In the infinite-width limit the NTK stays constant during training, so the network trains as a kernel method (Jacot, Gabriel, and Hongler, 2020). Chizat, Oyallon, and Bach (2020) show that this lazy regime is a matter of scale: multiplying a model by a factor α , with its initial output kept bounded, brings its training closer to that of its linearisation as α grows, while the mean-field factor $1/m$ of a two-layer network of width m keeps the dynamics non-degenerate as $m \rightarrow \infty$. Kumar et al. (2024) propose that grokking is a transition from lazy to rich, feature-learning dynamics, which arises when the top eigenvectors of the initial NTK are misaligned with the labels, the dataset is large enough to generalise from but not so large that the training loss tracks the test loss, and training starts in the lazy regime. Mohamadi et al. (2024) observe that the Frobenius distance of the empirical NTK from its initial value is negligible until the training data are fitted and large afterwards, and prove that with fewer than Cp^2 training pairs, for a constant C they do not give, no input-output permutation-equivariant kernel method has expected loss below half that of the zero predictor on modular addition. Lyu et al. (2024) prove that homogeneous networks trained with large initialisation and small weight decay stay at a kernel predictor for a long time and then move sharply to a min-norm or max-margin predictor. Atanasov, Bordelon, and Pehlevan (2021) describe silent alignment, in which the kernel’s eigenstructure evolves while the kernel is small and only its overall scale grows afterwards, under small initialisation and whitened data.

Kernel alignment and spectral bias. On a fixed kernel, gradient descent fits the components of the labels along the eigenvectors with large eigenvalues first (section 3.4). Networks learn low frequencies first (Rahaman et al., 2019), which Basri et al. (2019) explain for a two-layer network on the sphere through the eigenvalues of its NTK at each frequency, and Canatar, Bordelon, and Pehlevan (2021) relate the generalisation of kernel regression to the share of the target’s power in the top eigenfunctions. Outside the kernel regime, the top eigenfunctions of the NTK align with the target during training (Kopitkov and Indelman, 2020), the NTK changes rapidly in a short initial transient and at constant velocity afterwards (Fort et al., 2020), and the spectrum of the tangent kernel becomes more anisotropic while its centred alignment with the class labels rises (Baratin et al., 2021). Shan and Bordelon (2022) give a theory of this alignment for deep linear and two-layer ReLU networks, in which, with several outputs, the kernel of each output aligns with its own target. Kumar et al. (2024) also compute an alignment of the initial kernel with the labels, on the test set; their printed formula has no centring, and they evaluate it at initialisation only.

What sets the grokking time. Khanh et al. (2026) clamp the weight norm at a multiple ρ of a critical norm and find that the time to grok grows exponentially in ρ . Khanh (2026) show that under cross-entropy the norm acts on this delay through the scale of the logits, while under squared error its effect is about half as large and runs through a different route. Kim (2026) derive, for linear models trained with heavy-ball momentum β and weight decay, a grokking time that scales as $(1 - \beta)/(\eta\lambda)$, and observe this scaling on modular addition. Lewkowycz and Gur-Ari (2021) prove, assuming a conjecture on the correlation functions of wide networks, that for a k -homogeneous network in the infinite-width limit trained by gradient flow, L_2 regularisation makes the NTK evolve as $\frac{d}{dt}\Theta_t = -2(k - 1)\lambda\Theta_t$, and that under squared error its eigenvectors stay fixed while its eigenvalues decay; Kumar et al. (2024) cite this result, in an argument they call speculative, for weight decay moving networks out of the lazy regime. The evolution in that result is a rescaling of the kernel. Its proof starts from the gradient-flow equation of the kernel at finite width,

$$\frac{d\Theta_t(x, x')}{dt} = -2(k - 1)\lambda\Theta_t(x, x') + T_t(x, x') + T_t(x', x), \quad (1)$$

where $T_t(x, x') = -\sum_{x''} \partial_{\mu\nu} f_t(x) \partial_{\mu} f_t(x') \partial_{\nu} f_t(x'') \ell'(x'')$ sums over the training inputs and repeated parameter indices, and shows that T_0 has expectation of order n^{-1} in the width n , so that T drops out in the limit (Lewkowycz and Gur-Ari, 2021). In (1) the weight decay appears explicitly only in the term that rescales the kernel; T_t depends on it only through the trajectory. For homogeneous networks trained on the squared error with weight decay, Pracher et al. (2026) show that a residual persists after the training data are fitted, write the evolution of the kernel as a term driven by that residual plus a weight-decay term that shrinks the kernel, and predict a grokking time inversely proportional to the product of learning rate and weight decay, which they observe on modular addition; Xu, Vardi, and Safran (2026) prove bounds on the grokking time of over-parameterised linear regression under gradient descent with weight decay. Both mechanisms need weight decay, and none of these accounts separates the scale of the kernel from its shape.

3 Methods

3.1 Task, network and training

The task is the modular addition of Gromov (2023) at the baseline of Kumar et al. (2024). For a modulus p , each pair (a, b) of residues in $\{0, \dots, p-1\}$ is encoded as the concatenation $x_{a,b} \in \mathbb{R}^{2p}$ of the one-hot codes of a and b , and its target $y_{a,b} \in \mathbb{R}^p$ is the one-hot code of $(a+b) \bmod p$. The dataset holds all p^2 pairs. Each seed draws its own split of the pairs into a training set $\mathcal{T}$ and a test set, as well as its own initial weights.

The network is the two-layer perceptron without biases of Gromov (2023),

$$f(x; \theta) = c_N W^{(2)} \phi(D^{-1/2} W^{(1)} x), \quad (2)$$

with input dimension $D = 2p$, width N , weights $W^{(1)} \in \mathbb{R}^{N \times D}$ and $W^{(2)} \in \mathbb{R}^{p \times N}$ whose entries are drawn from $\mathcal{N}(0, 1)$, and an activation ϕ applied to each entry. Gromov (2023) derives his exact solution for the quadratic activation, finds his empirical results only mildly sensitive to the choice of activation, and reports that the same weights work for the rectifier at a larger width; we use the rectifier $\phi(h) = \max\{h, 0\}$. A network function is k -homogeneous if scaling every parameter by $c > 0$ scales it by c^k , and a fully connected network with L layers and rectifier activations is L -homogeneous (Lewkowycz and Gur-Ari, 2021), so (2) is 2-homogeneous; the centred predictor below, which subtracts the constant $f(x; \theta_0)$, is not. The readout factor is $c_N = N^{-1/2}$ in the NTK parameterisation of Jacot, Gabriel, and Hongler (2020) and $c_N = N^{-1}$ in the mean-field parameterisation of Gromov (2023). Following Kumar et al. (2024), the trained predictor is the centred and rescaled function

$$\tilde{f}(x; \theta) = \alpha [f(x; \theta) - f(x; \theta_0)], \quad (3)$$

where θ_0 holds the initial weights, and the learning rate is $\eta = \eta_0/\alpha^2$, which keeps the initial rate of change of the function the same for every α . The loss is the squared error averaged over the p outputs and the training pairs, $L(\theta) = \frac{1}{p|\mathcal{T}|} \sum_{(a,b) \in \mathcal{T}} \|\tilde{f}(x_{a,b}; \theta) - y_{a,b}\|^2$, and training is full-batch gradient descent with weight decay,

$$\theta_{k+1} = (1 - \eta\lambda) \theta_k - \eta \nabla L(\theta_k). \quad (4)$$

This is the decoupled weight decay of Kim (2026) with no momentum buffer, and for plain gradient descent coupled L_2 regularisation and decoupled weight decay coincide (Kim, 2026). The grid sets the product $\eta\lambda$ directly.

3.2 The empirical tangent kernel

For a scalar output $g(x; \theta)$ the tangent features are the gradients $\nabla_{\theta} g(x; \theta)$, and the tangent kernel is the inner product of the features of two inputs (Baratin et al., 2021; Jacot, Gabriel, and Hongler, 2020). We evaluate it on all $n = p^2$ pairs, so that at step t the kernel is the matrix $K_t \in \mathbb{R}^{n \times n}$ with entries

$$[K_t]_{ij} = \langle \nabla_{\theta} g(x_i; \theta_t), \nabla_{\theta} g(x_j; \theta_t) \rangle. \quad (5)$$

Two scalar outputs are measured: the sum of the logits of (2) divided by $\sqrt{p}$, whose kernel is the pseudo-NTK of Mohamadi, Bae, and Sutherland (2022) and is reported in the main text, and the first logit, which Mohamadi et al. (2024) use in place of the full kernel of a vector output. The kernel is evaluated at initialisation, on a logarithmically spaced grid of steps, and at every event step of section 3.5.

3.3 Scale, shape and alignment

For a set $\mathcal{S}$ of pairs, $K_t[\mathcal{S}]$ is the submatrix of K_t on those pairs and $Y[\mathcal{S}]$ the matrix whose rows are their targets; $\mathcal{T}$ holds the training pairs of a seed and $\mathcal{V}$ its test pairs. Following Cortes, Mohri, and Rostamizadeh (2024), the Frobenius product is $\langle A, B \rangle_F = \text{tr}(A^\top B)$ with norm $\|A\|_F = \sqrt{\langle A, A \rangle_F}$, and $K_c = CKC$ centres an $r \times r$ kernel matrix with $C = I_r - \frac{1}{r} \mathbf{1}\mathbf{1}^\top$. All four statistics are measured on the test pairs of a seed. The first three take the kernel $K_t = K_t[\mathcal{V}]$ of the summed logits; the alignment takes the tangent kernel Θ_t of all c logits on the same n pairs, an $nc \times nc$ matrix, and the vector $y \in \{0, 1\}^{nc}$ that stacks their one-hot targets $Y[\mathcal{V}]$ (Baratin et al., 2021, supplement, Sampled Versions). We record

$$\begin{aligned} S_t &= \frac{\|K_t\|_F}{\|K_0\|_F}, & R_t &= 1 - \frac{\text{tr}(K_t K_0^\top)}{\sqrt{\text{tr}(K_t K_t^\top)} \sqrt{\text{tr}(K_0 K_0^\top)}}, \\ D_t &= \frac{\|K_t - K_0\|_F}{\|K_0\|_F}, & A_t &= \text{CKA}(\Theta_t, yy^\top), & \text{CKA}(K, K') &= \frac{\text{tr}(K_c K'_c)}{\|K_c\|_F \|K'_c\|_F}. \end{aligned} \quad (6)$$

The scale S_t is the Frobenius norm of the kernel relative to its value at initialisation. Shan and Bordelon (2022) find that this norm grows during training, and in the silent alignment of Atanasov, Bordelon, and Pehlevan (2021) the kernel first aligns while small and then evolves in scale only, $K(x, x', t) = g(t)K_\infty(x, x')$. The shape change R_t is the kernel distance of Fort et al. (2020) between the kernels at step t and at initialisation, which compares two kernel matrices in a scale-invariant manner. The variation D_t is the total relative variation of the kernel that Geiger et al. (2020) measure on the test set to quantify feature training. The alignment A_t is the centred kernel alignment (CKA) of the tangent kernel with the class label kernel $yy^\top$ on a held-out set, as Baratin et al. (2021) measure it after Cortes, Mohri, and Rostamizadeh (2024); its normalisation makes it invariant under isotropic rescaling (Baratin et al., 2021). It takes all the logits rather than their sum because the sum’s kernel approximates the full one less well as training proceeds (Mohamadi, Bae, and Sutherland, 2022).

Because the first three statistics are taken on one matrix, the variation is a function of the scale and the shape,

$$D_t^2 = S_t^2 - 2S_t(1 - R_t) + 1, \quad (7)$$

and $R_t = 0$ exactly when $K_t = cK_0$ for some $c > 0$ (section A.1). A kernel that only rescales therefore has $D_t = |S_t - 1|$, which is as large as its change of scale, so a large D_t does not by itself show that the kernel changed shape.

3.4 Spectral measures

In the linear dynamics of Arora et al. (2019), the predictions u_k on the training pairs follow $u_{k+1} = u_k - \eta G(u_k - y)$ with a fixed Gram matrix G , from $u_0 = 0$. If G has eigenvalues ω_i^G with orthonormal eigenvectors g_i , the residual after k steps is $\|y - u_k\| = \sqrt{\sum_i (1 - \eta \omega_i^G)^{2k} (g_i^\top y)^2}$, so the part of the labels along an eigenvector with a large eigenvalue is fitted first; Jacot, Gabriel, and Hongler (2020) give the continuous-time form, in which the component along each kernel principal component decays exponentially at a rate set by its eigenvalue. Eigenvalues are written ω to keep them apart from the weight decay λ . The measures use the kernel on all $n = p^2$ pairs, which covers every input of the task. The centred kernel $CK_t C$, with $C = I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$, is symmetric and positive semi-definite (Cortes, Mohri, and Rostamizadeh, 2024), so it has orthonormal eigenvectors $v_1, \dots, v_n$ with eigenvalues $\omega_1 \geq \dots \geq \omega_n \geq 0$ (Axler, 2024). We record the target power along each eigenvector, $P_{t,i} = \sum_{c=1}^p (v_i^\top C y_c)^2$, where y_c is the c th column of Y , and its cumulative share over the leading eigenvectors, which Canatar, Bordelon, and Pehlevan (2021) call the cumulative power distribution and use to explain why a kernel generalises on one task and not on another. The centring follows Baratin et al. (2021), who report results for centred tangent features; Jacot, Gabriel, and Hongler (2020) note that their kernel principal component analysis is not centred, so its first component is almost the constant function. From the singular values $\sigma_1 \geq \dots \geq \sigma_n \geq 0$ of $CK_t C$ we compute the effective rank of Roy and Vetterli (2007), which Baratin et al. (2021) apply to the tangent kernel, and from the eigenvalues the trace ratios of Baratin et al. (2021),

$$\text{erank}_t = \exp\left(-\sum_j \mu_j \log \mu_j\right), \quad \mu_j = \frac{\sigma_j}{\sum_i \sigma_i}, \quad T_{t,k} = \frac{\sum_{j \leq k} \omega_j}{\sum_j \omega_j}, \quad (8)$$

which measure how evenly the kernel spreads over its eigenvectors; $T_{t,k}$ is the share of the trace carried by the k largest eigenvalues, which is how Baratin et al. (2021) describe their trace ratios: the relative importance of the top k eigenvalues. Roy and Vetterli (2007) take $0 \log 0 = 0$, so zero singular values, such as the one along the constant vector, which C maps to zero, do not enter the effective rank.

3.5 Event times

Following Khanh et al. (2026), the memorisation step of a seed is the first step at which its training accuracy reaches 0.99, and its grokking step at level ℓ is the first step at which its test accuracy reaches ℓ , for each $\ell \in \{0.80, 0.90 \text{ and } 0.95\}$, the test-accuracy thresholds at which Khanh et al. (2026) check that their critical weight norm does not depend on the threshold. A seed that has not reached a level within the step budget is kept, and its event time is recorded as censored at the budget.

Within a cell, the seeds give the Kaplan–Meier estimate of the probability that an event has not occurred by step t ,

$$\hat{F}(t) = \prod_{j|t_j \leq t} \frac{n_j - d_j}{n_j}, \quad (9)$$

where $t_1 < t_2 < \dots$ are the distinct steps at which the event occurs, d_j seeds reach it at t_j and n_j are still at risk there. Greenwood’s formula estimates its variance as $\hat{F}^2(t) \sum_{j|t_j \leq t} d_j / [n_j(n_j - d_j)]$, and with $\hat{s}^2(t) = \sum_{j|t_j \leq t} d_j / [n_j(n_j - d_j)] / [\log \hat{F}(t)]^2$ the approximate 95% interval $\hat{F}(t)^{\exp[\pm 1.96 \hat{s}(t)]}$ stays within $[0, 1]$ (Kalbfleisch and Prentice, 2002). The median event time of a cell is the step at which $\hat{F}$ falls to one half. A cell whose $\hat{F}$ stays above one half up to the budget has no median within it, and at a grokking level it does not grok.

Each run is also labelled by its phase at the last step, following Pracher et al. (2026): grokking when its training and test accuracies both end above a threshold, memorisation when only its training accuracy does, forgetting when its training accuracy rose above the threshold earlier and ends below it, and no fitting otherwise. The phases describe the cells and do not select runs. A run keeps its event times whatever its phase, since in the analysis of one failure mode a unit that fails by another mode, or is removed from test without failing, contributes its running time as a censoring time (Nelson, 1990).

Three censored regressions, fitted with lifelines (Davidson-Pilon, 2019), relate the event times to the settings of the cells. The log-normal accelerated failure time model takes the survival function of an event time T with covariates x to be

$$S(t | x) = 1 - \Phi\left(\frac{\log t - \mu(x)}{\sigma}\right), \quad \mu(x) = a_0 + \sum_i a_i x_i, \quad (10)$$

where Φ is the standard normal distribution function, so a unit increase in x_i multiplies the median event time by e^{a_i} . Its parameters are maximum likelihood estimates. With the step budget τ as the common censoring time, seed i contributes the log likelihood

$$\mathcal{L}_i = I_i \left[-\log \sigma - \frac{1}{2} \log(2\pi) - \frac{1}{2} z_i^2 \right] + (1 - I_i) \log[1 - \Phi(\zeta_i)] \quad (11)$$

in terms of its log time, where $z_i = (\log t_i - \mu(x_i))/\sigma$, $\zeta_i = (\log \tau - \mu(x_i))/\sigma$, and $I_i = 1$ if the event is observed and $I_i = 0$ otherwise (Nelson, 1990). This is the likelihood of Kalbfleisch and Prentice (2002) for a log-linear model under independent censoring, with normal errors. The additive model of Aalen (1989) takes the intensity of the event for run i at step t to be $\alpha_0(t) + \sum_j \alpha_j(t) Z_j^i$ while the run has not reached the level and zero afterwards, and estimates the cumulative regression functions $A_j(t) = \int_0^t \alpha_j(s) ds$ by

$$A^*(t) = \sum_{T_k \leq t} [Y(T_k)^\top Y(T_k)]^{-1} Y(T_k)^\top I_k, \quad (12)$$

where the event times $T_1 < T_2 < \dots$ are ordered, row i of $Y(t)$ is $(1, Z_1^i, \dots, Z_r^i)$ while run i is at risk and zero otherwise, and I_k is the indicator of the run whose event occurs at T_k . Unlike (10) with logarithmic covariates, (12) admits a covariate at zero, such as $\eta\lambda = 0$. The runs of one seed share its split and initial weights, so the intervals resample seeds, each with all its runs and their censoring, which keeps every group intact as the bootstrap for grouped and censored data does (Davison and Hinkley, 1997), and take the bias-corrected and accelerated interval (Efron and Tibshirani, 1994).

Each event and level is fitted separately. The covariates of (10) and (12) are the settings of a cell, fixed in time. As $\log(\eta\lambda)$ is undefined at $\eta\lambda = 0$, (10) is fitted twice: to the cells without weight decay, with indicators of the levels of α , and to the cells with $\eta\lambda > 0$, all at the α of the pilot cell, with the single covariate $\log(\eta\lambda)$, while (12) takes the pilot cell and the cells of the second stage together, $\eta\lambda = 0$ included, with $\eta\lambda$ itself. Every seed has one run in each of these cells, the balanced grouping for which resampling whole groups is justified (Davison and Hinkley, 1997, Sec. 3.8), while the scan trains a single seed. The estimator is defined only while $Y(t)$ has full rank, and the step at which the rank falls is a final censoring time (Aalen, 1989, Sec. 4.1); with indicators of α it would stop when the first group of runs at one α has left the risk set. Aalen (1989) states his results for covariates fixed in time, and lifelines fits (12) only for such covariates. The statistics of section 3.3 change along training, so they enter a third model, the relative risk model with time-dependent covariates of Kalbfleisch and Prentice (2002),

$$h(t | Z(t)) = h_0(t) \exp[Z(t)^\top \gamma], \quad (13)$$

whose coefficients maximise the partial likelihood

$$L(\gamma) = \prod_{j=1}^k \frac{\exp[Z_j(t_j)^\top \gamma]}{\sum_{l=1}^n Y_l(t_j) \exp[Z_l(t_j)^\top \gamma]}, \quad (14)$$

where $t_1 < \dots < t_k$ are the event times, Z_j belongs to the run whose event occurs at t_j , and $Y_l(t)$ is one while run l is at risk. Kalbfleisch and Prentice (2002) write it for untied event times; runs can reach an event at the same step, and the lifelines fitter handles such ties by Efron's method. Here $Z(t) = (S_t, R_t, A_t)$ is held from each checkpoint to the next, so its paths are left-continuous as the model requires. These covariates are internal: the run that reaches the event generates them. The model therefore describes the rate of the event given the current kernel and has no survivor function of its own. When a setting acts on the event through these covariates, an analysis conditional on them shows little effect of that setting (Kalbfleisch and Prentice, 2002).

3.6 Choice of levels

Where weight decay sets the grokking time, (10) with the single covariate $x = \log(\eta\lambda)$ makes $\log T$ normal with mean $a_0 + a_1x$ and variance σ^2 . Without censoring this is the simple linear regression model, whose designs Berger and Wong (2009) construct as follows. A design ξ places a proportion w_j of the runs at level x_j , $j = 1, \dots, m$, with $\sum_j w_j = 1$, on levels rescaled to $[-1, 1]$, and has the information matrix

$$M(\xi) = \sum_{j=1}^m w_j f(x_j) f(x_j)^\top, \quad f(x) = (1, x)^\top. \quad (15)$$

With N runs, $n_j = Nw_j$ of them at x_j , the least squares estimates have covariance $\sigma^2(X^\top X)^{-1}$, where $X^\top X = NM(\xi)$, and the slope has variance $\text{var}(\hat{a}_1) = \sigma^2 / \sum_i (x_i - \bar{x})^2$. The design that minimises this variance is unique and places half the runs at each end of the interval, and for the straight line it is also D-, A-, E- and G-optimal. When the line is in doubt, a quadratic term measures its curvature. The D-optimal design for the quadratic places a third of the runs at each of -1 , 0 and 1 , and the design for its curvature coefficient alone places half of them at 0 and a quarter at each end. For many models with p parameters an optimal design has between p and $p(p+1)/2$ distinct levels, and a design ξ is compared with the D-optimal design ξ^* of the model that holds by its relative efficiency

$$\text{RE}_D = \left\{ \frac{\det \text{Cov}(\hat{a}^*)}{\det \text{Cov}(\hat{a})} \right\}^{1/p}, \quad (16)$$

where $\hat{a}^*$ and $\hat{a}$ are the estimates under ξ^* and ξ .

Censoring at the budget changes the information a run carries. For $\mu(x) = a_0 + a_1x$ and censoring at $\log \tau$, a run at level x contributes to the parameters (a_0, a_1, σ) the expected information

$$F_x = \frac{1}{\sigma^2} \begin{pmatrix} A(\zeta) & xA(\zeta) & B(\zeta) \\ xA(\zeta) & x^2A(\zeta) & xB(\zeta) \\ B(\zeta) & xB(\zeta) & C(\zeta) \end{pmatrix}, \quad (17)$$

where $\zeta = (\log \tau - \mu(x))/\sigma$ and, with ϕ and Φ evaluated at ζ ,

$$\begin{aligned} A(\zeta) &= \Phi - \left[\zeta - \frac{\phi}{1 - \Phi} \right] \phi, & B(\zeta) &= -\phi \left[1 + \zeta \left(\zeta - \frac{\phi}{1 - \Phi} \right) \right], \\ C(\zeta) &= 2\Phi - \zeta\phi \left[1 + \zeta^2 - \frac{\zeta\phi}{1 - \Phi} \right]. \end{aligned} \quad (18)$$

The information of a plan is the sum of F_x over its runs, the covariance of the maximum likelihood estimates is its inverse, and a function h of the parameters has variance $HF^{-1}H^\top$, where H is the gradient of h (Nelson, 1990). Because ζ depends on the parameters, an optimum plan for censored data is found by numerical search for guessed values of them and is only locally optimum, and good compromise plans run more units at low stress, here small $\eta\lambda$, than at high (Nelson, 1990).

Across the level at which the grokking time returns to its value without weight decay, the mean log time is a broken line, $\mu(x) = a_0 + a_1x + b(x - \psi)_+$ with $(x - \psi)_+ = (x - \psi)I(x > \psi)$. The method of Muggeo (2003) applies to any regression model with a linear predictor, survival models included, so it runs inside the censored likelihood of (10). Step s of the fit sets $U^{(s)} = (x - \psi^{(s)})_+$ and $V^{(s)} = -I(x > \psi^{(s)})$, fits the linear predictor $a_1x + bU^{(s)} + \kappa V^{(s)}$, and updates $\psi^{(s+1)} = \psi^{(s)} + \hat{\kappa}/\hat{b}$ until $\hat{\kappa}$ vanishes, when every estimate, $\hat{\psi}$ included, is maximum likelihood. Fixing the left slope a_1 at zero gives a threshold model, in which the grokking time does not change below the break, and whether the left slope is zero is a Wald test of a_1 in the unconstrained fit (Muggeo, 2008). Whether a break exists at all is not a Wald test of b , since ψ vanishes under $b = 0$; the segmented package bounds the p -value of that test by the method of Davies (Muggeo, 2008). The break has the Wald interval $\hat{\psi} \pm 1.96 \text{SE}(\hat{\psi})$, whose standard error comes from the delta method for the ratio $\hat{\kappa}/\hat{b}$ and, at convergence, where $\hat{\kappa} = 0$, reduces to $\text{SE}(\hat{\kappa})/|\hat{b}|$ (Muggeo, 2003, 2008). When the break lies away from the middle of the range of x , its estimate is biased with an asymmetric distribution and the Wald interval can perform poorly, in part because $\text{SE}(\hat{\psi})$ may understate its variability (Muggeo, 2017). The break depends on constants that the grid itself estimates, so a design for it is also only locally optimal; a sequential design takes the guesses from a first stage, which here is a scan of weight decays, and updates the design with their estimates (Berger and Wong, 2009).

Park (1978) constructs D-optimal designs for segmented polynomials with a known break. For k distinct levels, $|X^\top X|$ is largest when every level has the same number of runs, and under continuity at the break the design is D-optimal on each segment with a level at the break. A line below the break and a quadratic above it therefore have the design $x \in \{-1, \psi, (1 + \psi)/2, 1\}$ on the rescaled axis. The quadratic is one degree above the line expected above the break, as Park (1978) advises when lack of fit is to be tested, with equal numbers of runs at each level. The design is locally optimal at the estimated break. The first stage is the column of the grid of Pracher et al. (2026) at the baseline’s learning rate, one run at each weight decay, and its last grokking and first other point bound the range above, as their boundary is placed. The broken line fitted to its runs below that bound estimates the break, and the second stage places the design at the estimate on the range from the column’s lowest point to the bound.

4 Experiments

4.1 The pilot cell

The pilot cell is the baseline of Kumar et al. (2024): modulus $p = 23$, a fraction 0.90 of the pairs for training, width $N = 100$ in the NTK parameterisation, $\alpha = 1$, $\eta_0 = 100$, $\eta\lambda = 0$ and a budget of 100 000 steps, for 12 seeds.

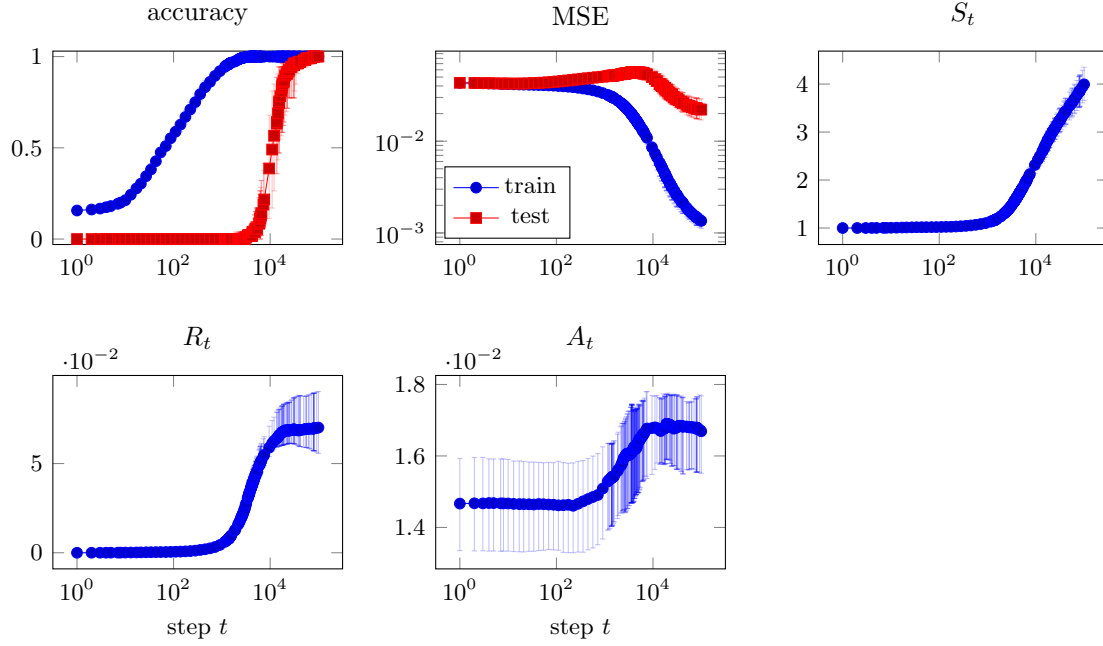


Figure 1: Train and test accuracy and MSE, the statistics S_t and R_t of the sum-over-logits kernel and the alignment A_t of the tangent kernel of all logits (6) against step t , at the baseline cell of Kumar et al. (2024). Points are medians over 12 seeds and error bars span their range.

Table 1: Steps at which the seeds of the pilot cell first reach each event: accuracy of training for memorisation and testing for grokking at levels following section 3.5. Median, minimum and maximum over seeds, and the number of seeds that reach the event within the budget.

Event	Level	Median	Minimum	Maximum	Reached
memorised	0.99	2500	1900	2800	12
grokked	0.80	17 000	11 000	32 000	12
grokked	0.90	21 000	15 000	38 000	12
grokked	0.95	31 000	18 000	57 000	12

Table 2: Statistics (6) and spectral measures (8) of the sum kernel of the pilot cell, with the alignment on the tangent kernel of all logits, at initialisation and at the last step, and the share of the centred target’s power on the $2(p - 1)$ leading eigenvectors. Minimum, median and maximum over seeds.

Statistic	Step	Minimum	Median	Maximum
A_t	0	0.013	0.015	0.016
A_t	100 000	0.016	0.017	0.018
D_t	0	0	0	0
D_t	100 000	2.7	3.1	3.4
R_t	0	0	0	0
R_t	100 000	0.056	0.070	0.090
S_t	0	1	1	1
S_t	100 000	3.7	4	4.4
erank_t	0	110	120	130
erank_t	100 000	49	53	54
target power on $v_1, \dots, v_{2(p-1)}$	0	0.0016	0.0018	0.0021
target power on $v_1, \dots, v_{2(p-1)}$	100 000	0.019	0.021	0.026
$T_{t,2(p-1)}$	0	0.76	0.77	0.78
$T_{t,2(p-1)}$	100 000	0.92	0.93	0.94

In Table 1 memorisation time is almost the same for every seed, between 1900 and 2800 steps, while the grokking event varies by a factor of about three across seeds at every level. The median step rises from 17 000 at level 0.80 to 31 000 at the 0.95 level. Hence the last fifteen percentage points of test accuracy take longer in general to gain than the first eighty.

The variation in the pilot cell is decomposed into its scale and shape terms using the identity (7). At the last step the scale in Table 2 is 4.0 and the shape is 0.070, so a kernel that had only rescaled would report a variation of 3.0 against the measured 3.1. In this case the scale term carries 94% of the squared variation at the final step, 92% at the grokking event and 70% at memorisation, as medians over seeds. Importantly during the run, the kernel grows by a factor of four while its shape changes by only a few percent. Fig. 1 also demonstrates how each term moves where the shape change seems to plateau by the grokking event, while the scale continues to grow through to the last step. The alignment rises from 0.015 at initialisation to 0.016 at memorisation and 0.017 at the grokking event, and holds that value to the last step, so the kernel moves a little towards the labels while it grows. Of the three covariates that enter the relative risk model of section 3.5, the scale is therefore the one that moves in this cell. Whether the event times follow the scale, or the shape takes more than one cell to arrive at an answer, section 4.3 explores this in more depth.

4.2 Moving one setting of the pilot cell

Each subsection below changes one setting of the pilot cell and holds all others fixed to investigate how grokking time is affected.

4.2.1 Output scale

Kumar et al. (2024) sweep α over $\{0.5, 1, 1.5, 2\}$ which we have repeated in Fig. 2 with twelve seeds per cell, keeping all other pilot cell parameters fixed. As expected, grokking arrives later

in lazier cells as α grows. At $\alpha = 1.5$ seven of the twelve seeds reach level 0.80 and none reaches 0.95 within 100 000 steps, and at $\alpha = 2$ no seed reaches any level.

Table 3: The sweep over α at the settings of the pilot cell, twelve seeds per cell: training accuracy is 0.99 and test accuracy is 0.95. Median, minimum and maximum are over the seeds that reached the event.

α	Memorised	Grokked			Reached
	Median	Median	Minimum	Maximum	
0.5	1400	5700	4100	7200	12
1	2500	31 000	18 000	57 000	12
1.5	3300	censored			0
2	4200	censored			0

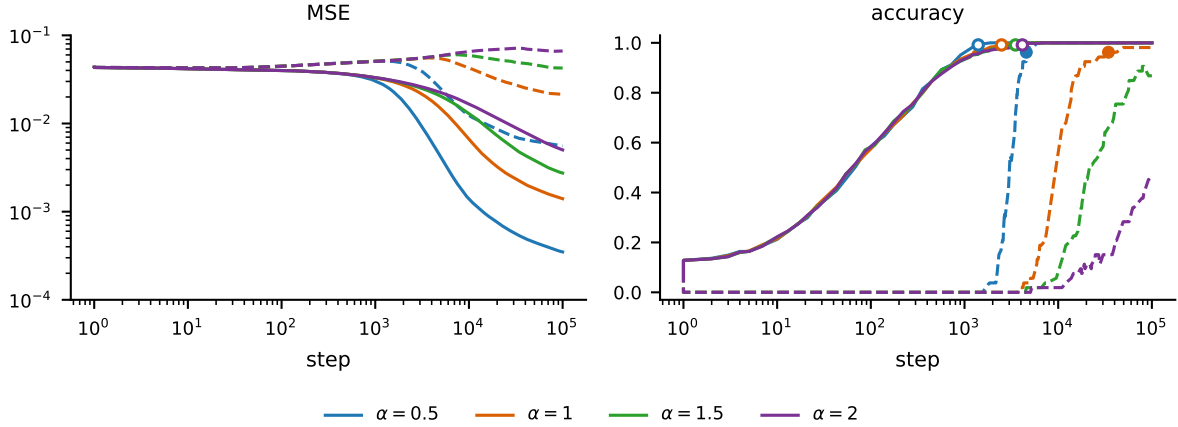


Figure 2: Seed 0 of the α sweep: MSE and accuracies against step. Solid lines are the training set and dashed lines the test set paths: hollow point is memorisation and solid point is grokking.

Fig. 2 also shows why the events are read from accuracy rather than from the loss. In the pilot cell the test loss plateaus between 0.017 and 0.029 while the test accuracy is 0.98 or 1, and a run of five times the budget only brings it to 0.013. At $\alpha = 1.5$ the seeds that reach level 0.80 do so with a test loss between 0.98 and 1.2 times its initial value, and at $\alpha = 2$ the loss ends above where it began while the accuracy has risen for all seeds. No single value of the loss marks generalisation, so the events are the accuracy levels of section 3.5. Kumar et al. (2024) report the same floor.

4.2.2 Weight decay

The grid of section 3.6 asks how far the weight decay can be raised at the settings of the pilot cell before it stops shortening the grokking time, and what happens at that edge. Every event in this subsection is at the grokking level 0.90, the level at which the design of section 3.6 was built and at which its figures are drawn; the other subsections use the level 0.95. The single-seed weight-decay sweep, the scan of section 3.6, trains one seed at each of 90 weight decays spaced evenly in $\ln \eta \lambda$ from 5×10^{-4} to 0.2, and labels each run by its phase at the last step (Figure 3). The phases are read at the last step against the threshold 0.9, so a run labelled grokking in

Figure 3 is one whose test accuracy ended above 0.9. Eight runs grok, every one of them at $\eta\lambda \leq 8.0 \times 10^{-4}$. Three runs memorise without grokking, between 8.6×10^{-4} and 9.8×10^{-4} . The remaining 79 runs, from 1.05×10^{-3} upwards, never fit the training data. The edge is therefore narrow: the weight decay can rise by a factor of 1.6 above the lowest point of the single-seed sweep before the network stops grokking, and by a further 1.2 before it stops fitting at all.

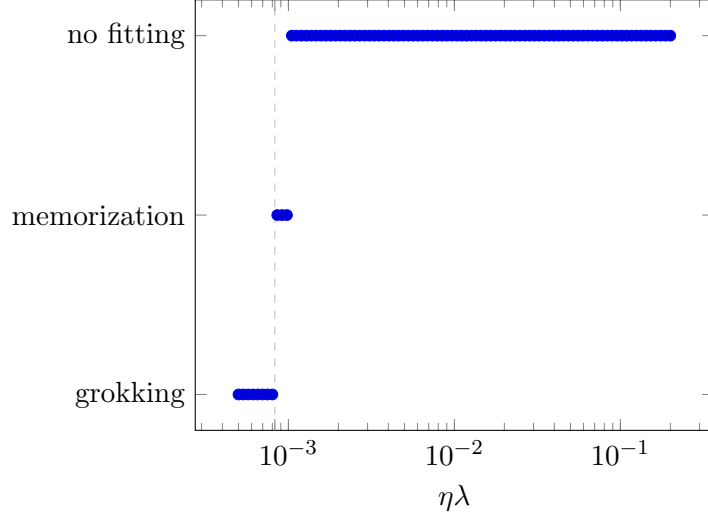


Figure 3: Outcome of each weight decay run at the last step against $\eta\lambda$. The dashed line is the upper end of the range in Table 4.

Table 4: The sequential design (Berger and Wong, 2009, Sec. 5.4). The single-seed sweep’s last grokking point, its first other point and their geometric midpoint, the upper end of the range (Pracher et al., 2026, App. G.2.1); the join $\hat{\psi}$ of the broken line fitted to the single-seed sweep’s runs, with its interval (Muggeo, 2008); the lower end of the single-seed sweep; and the four levels of Park (1978, Sec. 4.1) at x , $\ln \eta\lambda$ rescaled to $[-1, 1]$.

Quantity	$\eta\lambda$	x
last grokking point of the single-seed sweep	0.000 80	
first other point of the single-seed sweep	0.000 86	
upper end	0.000 83	
join, lower limit	0.000 72	
join $\hat{\psi}$	0.000 74	
join, upper limit	0.000 76	
lower end	0.000 50	
level 1	0.000 50	−1
level 2	0.000 74	0.55
level 3	0.000 78	0.78
level 4	0.000 83	1

Below the edge the grokking time rises steeply with the weight decay, and the broken line of section 3.6 describes the rise. Fitted to the grokking runs of the single-seed sweep at level 0.90, it places the break at $\eta\lambda = 7.4 \times 10^{-4}$, and consequently the twelve-seed sweep is placed around that estimate. Once the twelve-seed sweep had been trained, the broken line was fitted a second

time, to the runs of both weight-decay sweeps that lie within the range of the design (Figure 4). This second fit moves the break to 7.0×10^{-4} , with the interval 6.8×10^{-4} to 7.3×10^{-4} . Below the break the grokking time does not change with the weight decay according to the line of best fit. Above it the time doubles for every four percent increase in the weight decay, a change of slope b of 19 in log time per unit of $\ln \eta\lambda$.

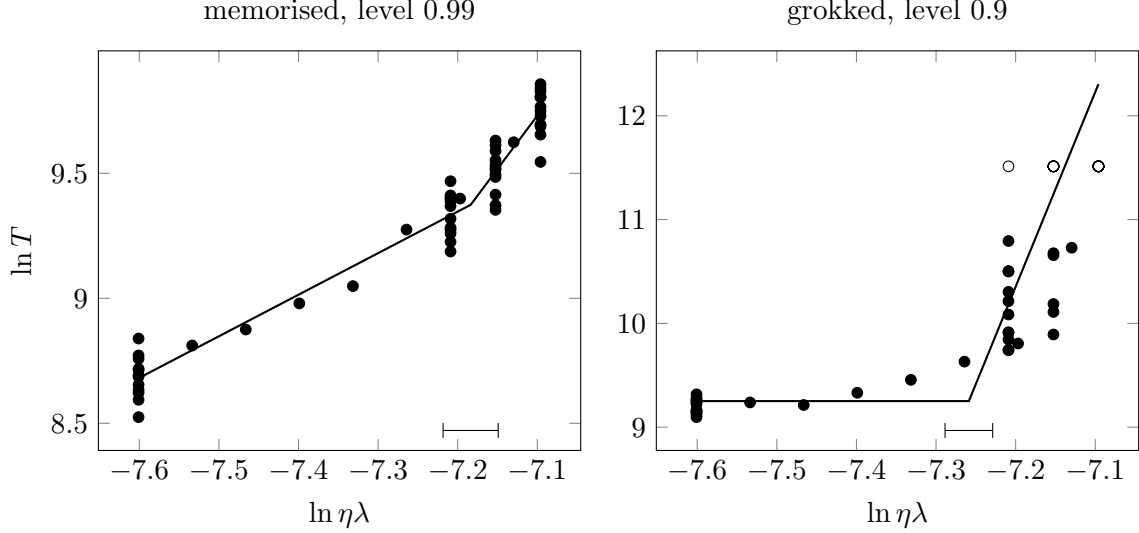


Figure 4: Log event time against $\ln \eta\lambda$ for the runs of the design, with the broken line of Muggeo (2003) chosen by AIC between a free and a zero left slope, and its join $\hat{\psi}$ with the delta-method interval drawn as a bar, as in Muggeo (2008, Fig. 1). Filled marks are observed events and open marks runs censored at the budget.

The four levels of Table 4 were trained with twelve seeds each (Figure 5). At $\eta\lambda = 5.0 \times 10^{-4}$ every seed groks, with a median grokking step of 9900 against 21 000 in the pilot cell. Thus the weight decay shortens the grokking time by a factor of 2.1. At 7.4×10^{-4} eleven of the twelve seeds grok within the budget, with a median of 24 000, at 7.83×10^{-4} five, and at 8.28×10^{-4} none. At the level 0.95 of the other subsections the same cells grok in twelve, six, two and no seeds, so the last five percentage points of test accuracy are what the edge costs. The memorisation step moves the other way, from 2500 in the pilot cell to 5900 at the lowest level and 17 000 at the highest, where every seed still memorises. The first-stage panel of Figure 5 is the sweep of Table 3 as a survival curve.

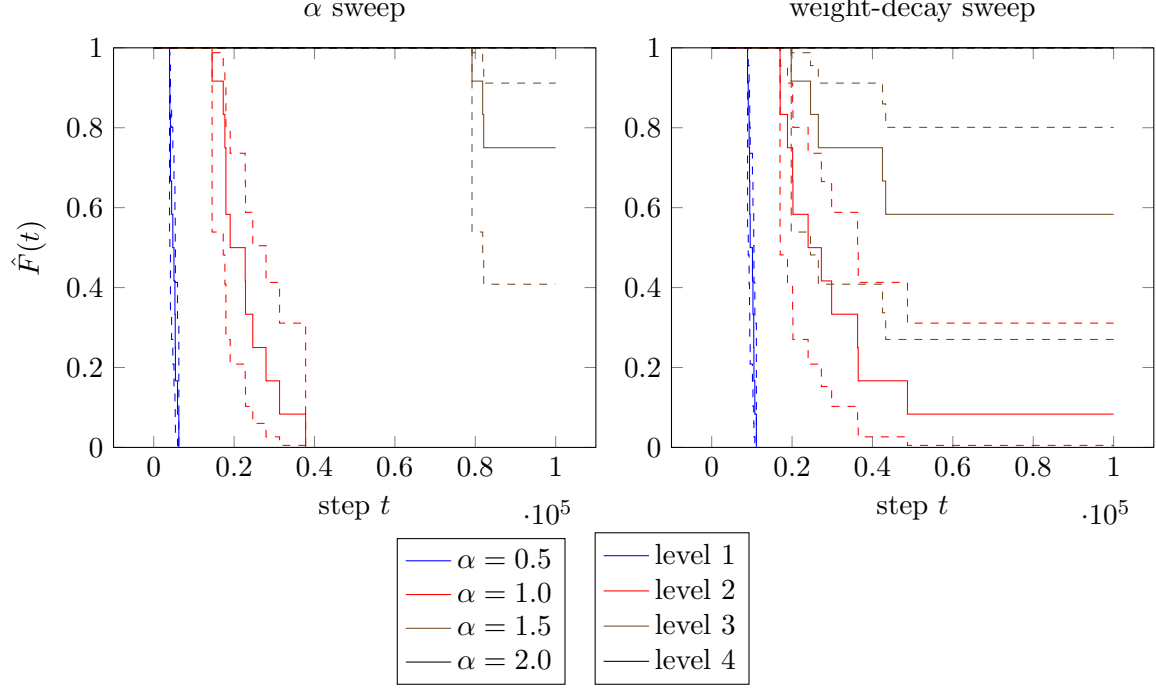


Figure 5: Kaplan–Meier estimates of the probability of not yet having grokked at level 0.9, by output scale in the alpha sweep and by weight decay level in the twelve-seed weight-decay sweep, with Greenwood’s exponential interval dashed (Kalbfleisch and Prentice, 2002, Sec. 1.4.1).

Table 5 gives the two arms of the accelerated failure time model (10), the output scale arm and the weight decay arm. The weight decay arm is fitted to the twelve-seed sweep, which lies above the break, and its coefficient of 5.0 on $\ln \eta \lambda$ at level 0.90, with the interval 3.9 to 6.1, is the steepness of that rise. In this range more weight decay delays grokking. The additive model (12), fitted to the pilot cell and the twelve-seed sweep together, gives the same sign (Figure 8). Neither fit says anything about weight decays below 5×10^{-4} , where the time falls.

Table 5: Coefficients of the log-normal accelerated failure time model (10) with their 95 % intervals, for the output scale arm against $\alpha = 1.0$ and the weight decay arm on $\ln \eta \lambda$.

Arm	Event	Level	Covariate	Coefficient	Lower	Upper
alpha	memorised	0.99	$\alpha = 0.5$	−0.55	−0.65	−0.45
alpha	memorised	0.99	$\alpha = 1.5$	0.31	0.22	0.41
alpha	memorised	0.99	$\alpha = 2.0$	0.56	0.46	0.66
alpha	grokked	0.80	$\alpha = 0.5$	−1.5	−1.7	−1.3
alpha	grokked	0.80	$\alpha = 1.5$	1.6	1.3	1.8
alpha	grokked	0.90	$\alpha = 0.5$	−1.5	−1.7	−1.3
alpha	grokked	0.90	$\alpha = 1.5$	1.6	1.4	1.9
alpha	grokked	0.95	$\alpha = 0.5$	−1.7	−2	−1.5
decay	memorised	0.99	$\ln \eta \lambda$	2	1.8	2.1
decay	grokked	0.80	$\ln \eta \lambda$	3.4	2.6	4.2
decay	grokked	0.90	$\ln \eta \lambda$	5	3.9	6.1
decay	grokked	0.95	$\ln \eta \lambda$	6.1	4.8	7.3

4.3 The kernel at the grokking event

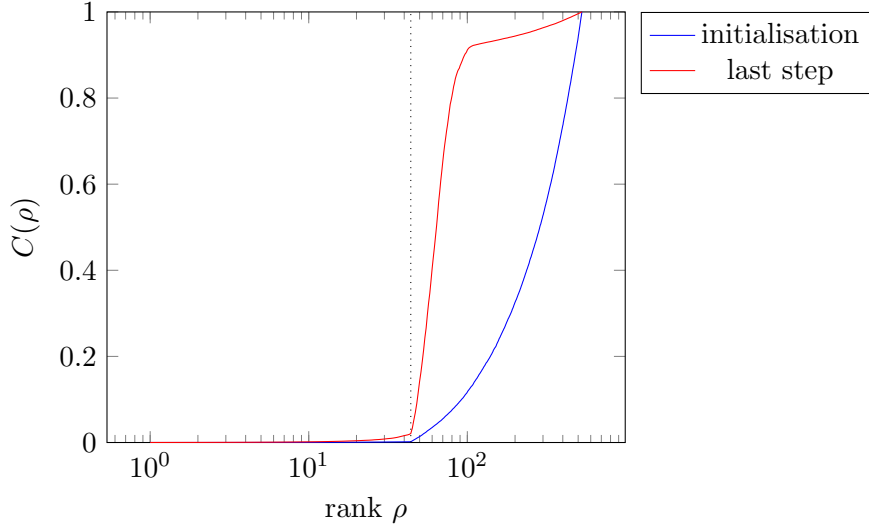


Figure 6: Cumulative share $C(\rho)$ of the centred target’s power on the leading ρ eigenvectors of the sum kernel of the pilot cell, at initialisation and at the last step, the curve of Canatar, Bordelon, and Pehlevan (2021, Eq. 6, Fig. 1e). Medians over 12 seeds; the dotted line marks $\rho = 2(p - 1)$.

Table 6: Coefficients of the relative risk model with time-dependent covariates (Kalbfleisch and Prentice, 2002, Eq. 6.14) on S_t , R_t and A_t (6), with their 95 % intervals and p -values.

Event	Level	Covariate	Coefficient	Lower	Upper	p
memorised	0.99	A_t	670	330	1000	0.000 10
memorised	0.99	R_t	38	4	72	0.029
memorised	0.99	S_t	9.4	7.5	11	2.6×10^{-22}
grokked	0.80	A_t	1200	830	1500	3.9×10^{-12}
grokked	0.80	R_t	16	−14	46	0.29
grokked	0.80	S_t	0.89	0.34	1.4	0.0015
grokked	0.90	A_t	1100	720	1500	7.8×10^{-9}
grokked	0.90	R_t	21	−13	55	0.22
grokked	0.90	S_t	1	0.41	1.6	0.000 90
grokked	0.95	A_t	1300	830	1700	3.2×10^{-8}
grokked	0.95	R_t	17	−20	54	0.36
grokked	0.95	S_t	1.3	0.60	1.9	0.000 19

4.4 What times the event

FiXme Note: What times the event: to come.

5 Conclusion

FiXme Fatal: Conclusion: what the measurements show, and the limitations.

Contribution

FiXme Fatal: Contribution of each group member, from the group.

Acknowledgement

FiXme Fatal: Acknowledgement, written by the group.

A Derivations

A.1 Scale, shape and variation

Throughout, K_t and K_0 are real $r \times r$ kernel matrices on the test pairs of a seed with $K_0 \neq 0$ and $K_t \neq 0$, so that (6) is defined.

Results used. On the linear maps between two finite-dimensional inner product spaces, $\langle S, T \rangle = \text{tr}(T^* S)$ is an inner product, which with respect to orthonormal bases is the standard inner product of the matrix entries, and its norm is the Frobenius norm (Axler, 2024). For real matrices the adjoint is the transpose, so this is the Frobenius product $\langle A, B \rangle_F = \text{tr}(A^\top B)$ of Cortes, Mohri, and Rostamizadeh (2024). Square matrices satisfy $\text{tr}(AB) = \text{tr}(BA)$ (Axler, 2024). In an inner product space,

$$\|u - v\|^2 = \|u\|^2 + \|v\|^2 - \langle u, v \rangle - \langle v, u \rangle,$$

over the real numbers $\langle u, v \rangle = \langle v, u \rangle$, and $\langle u, v \rangle = \|u\| \|v\|$ holds if and only if one of u and v is a nonnegative real multiple of the other (Axler, 2024).

The identity. By $\text{tr}(AB) = \text{tr}(BA)$, $\text{tr}(K_t K_0^\top) = \text{tr}(K_0^\top K_t) = \langle K_0, K_t \rangle_F$, and likewise $\text{tr}(K_t K_t^\top) = \|K_t\|_F^2$, so (6) gives

$$1 - R_t = \frac{\langle K_t, K_0 \rangle_F}{\|K_t\|_F \|K_0\|_F}.$$

The expansion above with the symmetry of the real inner product gives

$$\|K_t - K_0\|_F^2 = \|K_t\|_F^2 + \|K_0\|_F^2 - 2\langle K_t, K_0 \rangle_F.$$

Dividing by $\|K_0\|_F^2$, the left side is D_t^2 , the first term is S_t^2 , and

$$\frac{\langle K_t, K_0 \rangle_F}{\|K_0\|_F^2} = \frac{\|K_t\|_F}{\|K_0\|_F} \cdot \frac{\langle K_t, K_0 \rangle_F}{\|K_t\|_F \|K_0\|_F} = S_t(1 - R_t),$$

which gives (7).

When the shape is unchanged. By the expression for $1 - R_t$ above, $R_t = 0$ exactly when $\langle K_t, K_0 \rangle_F = \|K_t\|_F \|K_0\|_F$, that is, by the last result above, exactly when one of K_t and K_0 is a nonnegative real multiple of the other. Neither is zero, so the multiple is positive, and $R_t = 0$ exactly when $K_t = cK_0$ for some $c > 0$. Then $S_t = c$, and (7) gives $D_t^2 = (S_t - 1)^2$.

B Model checks

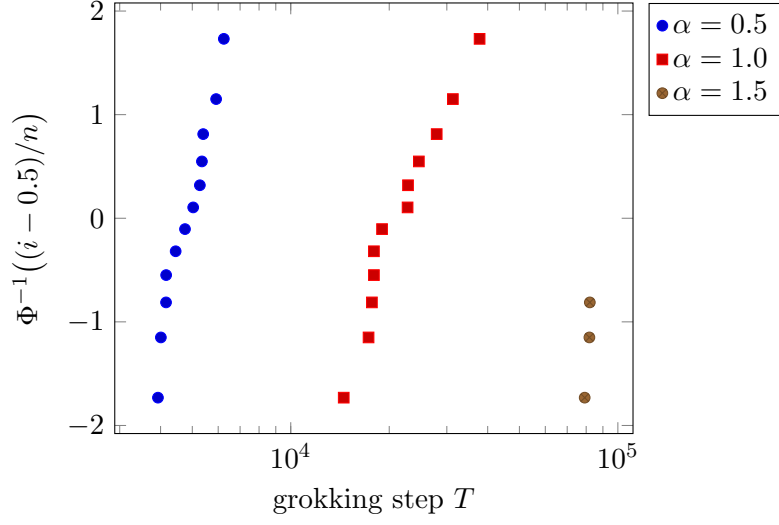


Figure 7: Log-normal probability plot of the grokking times at level 0.9 for each output scale of the first stage, each observed time at the normal quantile of its plotting position $(i - 0.5)/n$ and runs censored at the budget left out (Nelson, 1990, Ch. 3, Sec. 4.2). Parallel straight lines support the log-normal model (10).

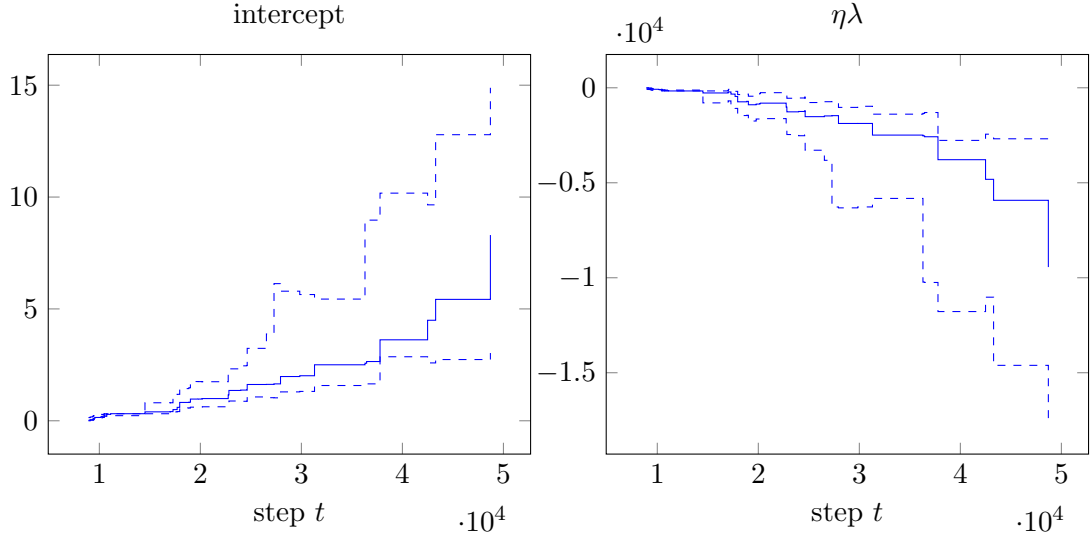


Figure 8: Cumulative regression functions of Aalen (1989) for grokking at level 0.9, with the bias-corrected and accelerated bootstrap interval over seeds dashed, drawn as in Aalen (1989, Fig. 7). A slope that changes along t is an effect that changes with time.

References

- Aalen, O. O. (Aug. 1989). “A Linear Regression Model for the Analysis of Life Times”. In: *Statistics in Medicine* 8.8, pp. 907–925. ISSN: 0277-6715, 1097-0258. DOI: 10.1002/sim.4780080803. (Visited on 09/28/2026).
- Arora, S. et al. (2019). *Fine-Grained Analysis of Optimization and Generalization for Overparameterized Two-Layer Neural Networks*. DOI: 10.48550/ARXIV.1901.08584.
- Atanasov, A., B. Bordelon, and C. Pehlevan (Dec. 2021). *Neural Networks as Kernel Learners: The Silent Alignment Effect*. DOI: 10.48550/arXiv.2111.00034. arXiv: 2111.00034 [stat.ML]. (Visited on 09/24/2026).
- Axler, S. (2024). *Linear Algebra Done Right*. Undergraduate Texts in Mathematics. Cham: Springer International Publishing. ISBN: 978-3-031-41025-3 978-3-031-41026-0. DOI: 10.1007/978-3-031-41026-0. (Visited on 09/28/2026).
- Baratin, A. et al. (Mar. 2021). *Implicit Regularization via Neural Feature Alignment*. DOI: 10.48550/arXiv.2008.00938. arXiv: 2008.00938 [cs.LG]. (Visited on 09/24/2026).
- Basri, R. et al. (2019). “The Convergence Rate of Neural Networks for Learned Functions of Different Frequencies”. In: DOI: 10.48550/ARXIV.1906.00425. (Visited on 09/28/2026).
- Berger, M. P. F. and W. K. Wong (May 2009). *An Introduction to Optimal Designs for Social and Biomedical Research*. 1st ed. Wiley. ISBN: 978-0-470-69450-3 978-0-470-74691-2. DOI: 10.1002/9780470746912. (Visited on 09/29/2026).
- Canatar, A., B. Bordelon, and C. Pehlevan (May 2021). “Spectral Bias and Task-Model Alignment Explain Generalization in Kernel Regression and Infinitely Wide Neural Networks”. In: *Nature Communications* 12.1, p. 2914. ISSN: 2041-1723. DOI: 10.1038/s41467-021-23103-1. arXiv: 2006.13198 [stat.ML]. (Visited on 09/24/2026).
- Chizat, L., E. Oyallon, and F. Bach (Jan. 2020). *On Lazy Training in Differentiable Programming*. DOI: 10.48550/arXiv.1812.07956. arXiv: 1812.07956 [math.OC]. (Visited on 09/24/2026).
- Cortes, C., M. Mohri, and A. Rostamizadeh (Apr. 2024). *Algorithms for Learning Kernels Based on Centered Alignment*. DOI: 10.48550/arXiv.1203.0550. arXiv: 1203.0550 [cs.LG]. (Visited on 09/24/2026).
- Davidson-Pilon, C. (Aug. 2019). “Lifelines: Survival Analysis in Python”. In: *Journal of Open Source Software* 4.40, p. 1317. ISSN: 2475-9066. DOI: 10.21105/joss.01317. (Visited on 09/24/2026).
- Davison, A. C. and D. V. Hinkley (Oct. 1997). *Bootstrap Methods and Their Application*. 1st ed. Cambridge University Press. ISBN: 978-0-521-57391-7 978-0-521-57471-6 978-0-511-80284-3. DOI: 10.1017/CB09780511802843. (Visited on 09/24/2026).
- Efron, B. and R. Tibshirani (May 1994). *An Introduction to the Bootstrap*. 0th ed. Chapman and Hall/CRC. ISBN: 978-0-429-24659-3. DOI: 10.1201/9780429246593. (Visited on 09/24/2026).
- Fort, S. et al. (Oct. 2020). *Deep Learning versus Kernel Learning: An Empirical Study of Loss Landscape Geometry and the Time Evolution of the Neural Tangent Kernel*. DOI: 10.48550/arXiv.2010.15110. arXiv: 2010.15110 [cs.LG]. (Visited on 09/24/2026).
- Geiger, M. et al. (Nov. 2020). “Disentangling Feature and Lazy Training in Deep Neural Networks”. In: *Journal of Statistical Mechanics: Theory and Experiment* 2020.11, p. 113301. ISSN: 1742-5468. DOI: 10.1088/1742-5468/abc4de. (Visited on 09/24/2026).
- Gromov, A. (Jan. 2023). *Grokking Modular Arithmetic*. DOI: 10.48550/arXiv.2301.02679. arXiv: 2301.02679 [cs.LG]. (Visited on 09/24/2026).
- Jacot, A., F. Gabriel, and C. Hongler (Feb. 2020). *Neural Tangent Kernel: Convergence and Generalization in Neural Networks*. DOI: 10.48550/arXiv.1806.07572. arXiv: 1806.07572 [cs.LG]. (Visited on 09/24/2026).

- Kalbfleisch, J. D. and R. L. Prentice (Aug. 2002). *The Statistical Analysis of Failure Time Data*. 1st ed. Wiley Series in Probability and Statistics. Wiley. ISBN: 978-0-471-36357-6 978-1-118-03298-5. DOI: 10.1002/9781118032985. (Visited on 09/28/2026).
- Khanh, T. X. (June 2026). *What Does the Weight Norm Control in Grokking? Logit-scale Mediation under Cross-Entropy*. DOI: 10.48550/arXiv.2606.18465. arXiv: 2606.18465 [cs.LG]. (Visited on 09/24/2026).
- Khanh, T. X. et al. (June 2026). *The Weight Norm Sets the Grokking Timescale: A Causal Delay Law*. DOI: 10.48550/arXiv.2606.13753. arXiv: 2606.13753 [cs.LG]. (Visited on 09/24/2026).
- Kim, T. (July 2026). *Grokking on the Weight-Decay Clock: A Rate Hierarchy from Softly Broken Symmetries*. DOI: 10.48550/arXiv.2607.23967. arXiv: 2607.23967 [cs.AI]. (Visited on 09/24/2026).
- Kopitkov, D. and V. Indelman (Apr. 2020). *Neural Spectrum Alignment: Empirical Study*. DOI: 10.48550/arXiv.1910.08720. arXiv: 1910.08720 [cs.LG]. (Visited on 09/24/2026).
- Kumar, T. et al. (Apr. 2024). *Grokking as the Transition from Lazy to Rich Training Dynamics*. DOI: 10.48550/arXiv.2310.06110. arXiv: 2310.06110 [stat.ML]. (Visited on 09/24/2026).
- Lewkowycz, A. and G. Gur-Ari (Jan. 2021). *On the Training Dynamics of Deep Networks with ℓ_2 Regularization*. DOI: 10.48550/arXiv.2006.08643. arXiv: 2006.08643 [stat.ML]. (Visited on 09/24/2026).
- Liu, Z., E. J. Michaud, and M. Tegmark (Mar. 2023). *Omnigrok: Grokking beyond Algorithmic Data*. DOI: 10.48550/arXiv.2210.01117. arXiv: 2210.01117 [cs.LG]. (Visited on 09/24/2026).
- Lyu, K. et al. (Apr. 2024). *Dichotomy of Early and Late Phase Implicit Biases Can Provably Induce Grokking*. DOI: 10.48550/arXiv.2311.18817. arXiv: 2311.18817 [cs.LG]. (Visited on 09/24/2026).
- Mohamadi, M. A., W. Bae, and D. J. Sutherland (2022). “A Fast, Well-Founded Approximation to the Empirical Neural Tangent Kernel”. In: DOI: 10.48550/arXiv.2206.12543. arXiv: 2206.12543 [stat.ML]. (Visited on 09/24/2026).
- Mohamadi, M. A. et al. (July 2024). *Why Do You Grok? A Theoretical Analysis of Grokking Modular Addition*. DOI: 10.48550/arXiv.2407.12332. arXiv: 2407.12332 [cs.LG]. (Visited on 09/24/2026).
- Muggeo, V. M. R. (Oct. 2003). “Estimating Regression Models with Unknown Break-points”. In: *Statistics in Medicine* 22.19, pp. 3055–3071. ISSN: 0277-6715, 1097-0258. DOI: 10.1002/sim.1545. (Visited on 09/29/2026).
- (May 2008). “Segmented: An R Package to Fit Regression Models with Broken-Line Relationships”. In: *R News* 8.1, pp. 20–25. ISSN: 1609-3631. (Visited on 09/29/2026).
- Muggeo, V. M. (Sept. 2017). “Interval Estimation for the Breakpoint in Segmented Regression: A Smoothed Score-based Approach”. In: *Australian & New Zealand Journal of Statistics* 59.3, pp. 311–322. ISSN: 1369-1473, 1467-842X. DOI: 10.1111/anzs.12200. (Visited on 09/29/2026).
- Nanda, N. et al. (Oct. 2023). *Progress Measures for Grokking via Mechanistic Interpretability*. DOI: 10.48550/arXiv.2301.05217. arXiv: 2301.05217 [cs.LG]. (Visited on 09/24/2026).
- Nelson, W. (Feb. 1990). *Accelerated Testing: Statistical Models, Test Plans, and Data Analysis*. 1st ed. Wiley Series in Probability and Statistics. Wiley. ISBN: 978-0-471-52277-5 978-0-470-31679-5. DOI: 10.1002/9780470316795. (Visited on 09/29/2026).
- Park, S. H. (1978). “Experimental Designs for Fitting Segmented Polynomial Regression Models”. In: *Technometrics* 20.2, pp. 151–154. ISSN: 0040-1706. DOI: 10.2307/1268706. JSTOR: 1268706. (Visited on 09/29/2026).

- Power, A. et al. (Jan. 2022). *Grokking: Generalization beyond Overfitting on Small Algorithmic Datasets*. DOI: 10.48550/arXiv.2201.02177. arXiv: 2201.02177 [cs.LG]. (Visited on 09/24/2026).
- Pracher, L. et al. (2026). *A Spectral Theory of Grokking: Weight Decay Induces Feature Learning*. DOI: 10.48550/ARXIV.2609.26679.
- Rahaman, N. et al. (May 2019). *On the Spectral Bias of Neural Networks*. DOI: 10.48550/arXiv.1806.08734. arXiv: 1806.08734 [stat.ML]. (Visited on 09/24/2026).
- Roy, O. and M. Vetterli (Sept. 2007). “The Effective Rank: A Measure of Effective Dimensionality”. In: *15th European Signal Processing Conference (EUSIPCO 2007)*. Poznan, Poland: Zenodo, pp. 606–610. DOI: 10.5281/ZENODO.40328. (Visited on 09/24/2026).
- Shan, H. and B. Bordon (Feb. 2022). *A Theory of Neural Tangent Kernel Alignment and Its Influence on Training*. DOI: 10.48550/arXiv.2105.14301. arXiv: 2105.14301 [stat.ML]. (Visited on 09/24/2026).
- Xu, M., G. Vardi, and I. Safran (2026). *To Grok Grokking: Provable Grokking in Ridge Regression*. DOI: 10.48550/ARXIV.2601.19791.

Fixme Fatal: The group keeps one of the two consent statements below.

We give consent for this to be used as a teaching resource.
We DO NOT give consent for this to be used as a teaching resource.